

3 . S E M A N T I C S E G M E N T A T I O N A N D S T Y L E T R A N S F E R

A N T O N I O D Í A Z - C A N O

A J D I A Z C A N O @ C O M I L L A S . E D U

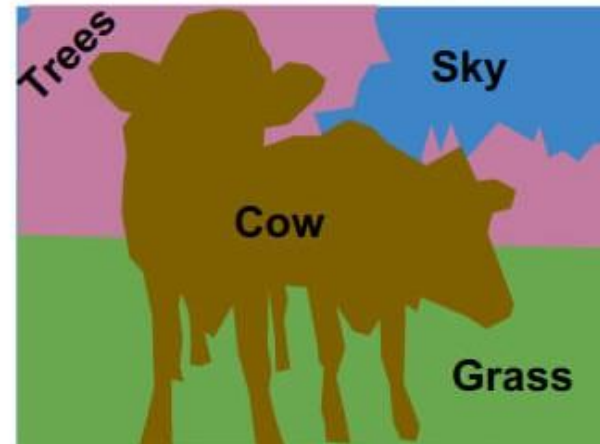
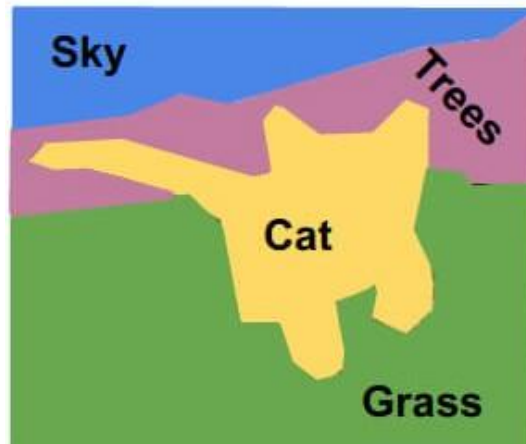
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1. Semantic Segmentation
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1 . S E M A N T I C S E G M E N T A T I O N

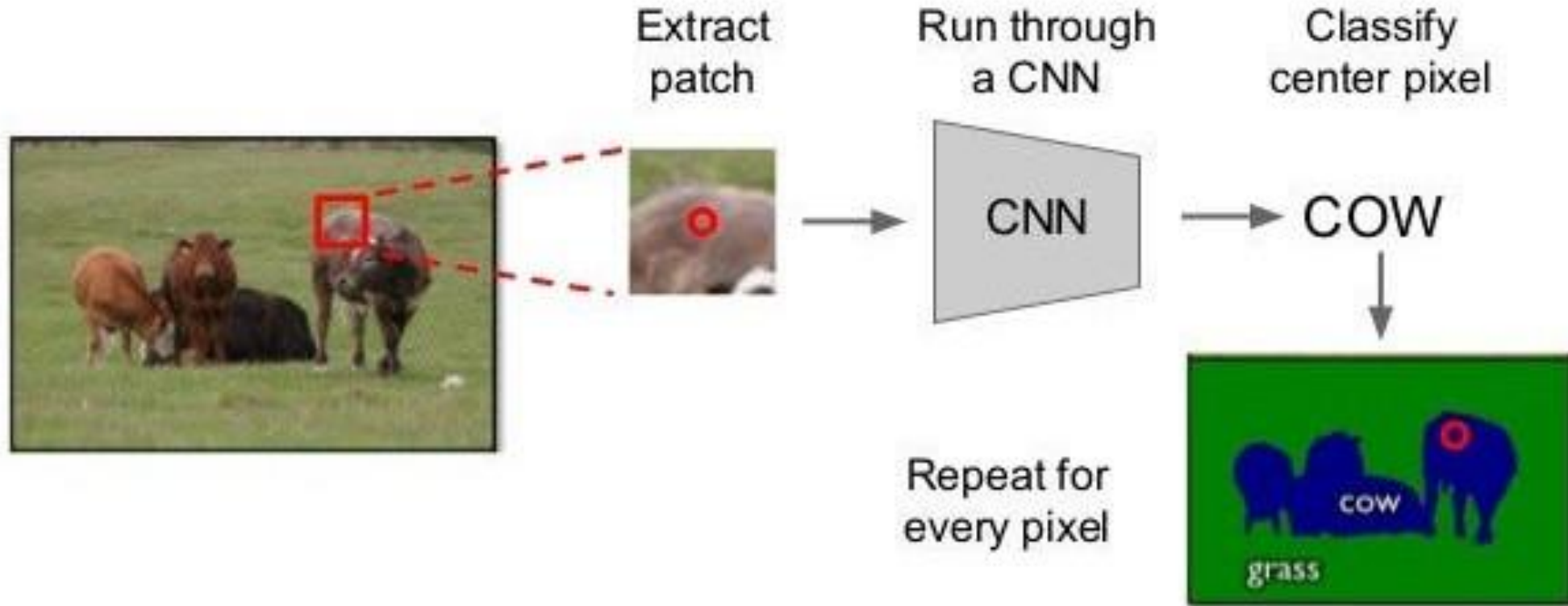
DEFINITION OF THE PROBLEM



Labeling is at pixel level!



NAÏVE APPROACH



Larger patches: + context, - precision for the center pixel.

Smaller patches: - context, + precision for the center pixel.

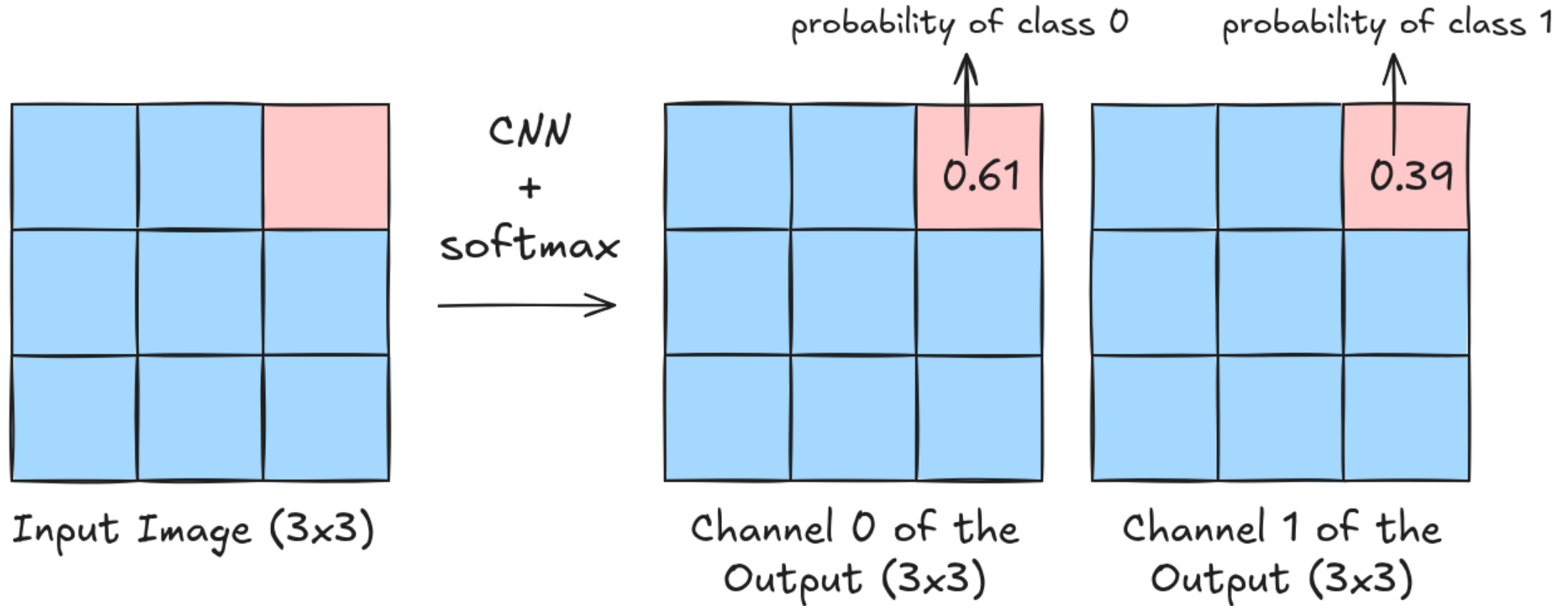
NAÏVE APPROACH

Problem: Too much computation needed and no features.

Solution: Let's try to classify all pixels at the same time!



POSSIBLE APPROACH



TRANSPPOSED CONVOLUTION

$$\begin{aligned}
 \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix} *^{-1} \begin{bmatrix} 0 & 1 \\ 2 & 3 \end{bmatrix} &= 0 \begin{bmatrix} 0 & 1 & - \\ 2 & 3 & - \\ - & - & - \end{bmatrix} + 1 \begin{bmatrix} - & 0 & 1 \\ - & 2 & 3 \\ - & - & - \end{bmatrix} + 2 \begin{bmatrix} - & - & - \\ 0 & 1 & - \\ 2 & 3 & - \end{bmatrix} + 3 \begin{bmatrix} - & - & - \\ - & 0 & 1 \\ - & 2 & 3 \end{bmatrix} \\
 &= \begin{bmatrix} 0 & 0 & - \\ 0 & 0 & - \\ - & - & - \end{bmatrix} + \begin{bmatrix} - & 0 & 1 \\ - & 2 & 3 \\ - & - & - \end{bmatrix} + \begin{bmatrix} - & - & - \\ 0 & 2 & - \\ 4 & 6 & - \end{bmatrix} + \begin{bmatrix} - & - & - \\ - & 0 & 3 \\ - & 6 & 9 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 4 & 6 \\ 4 & 12 & 9 \end{bmatrix}
 \end{aligned}$$



PADDING

Padding n x m consists just of removing the first and last n rows and m columns.

$$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 4 & 6 \\ 4 & 12 & 9 \end{pmatrix} = (4)$$

Padding 1x1 for the previous example.



STRIDE

Stride n consists just of including n "imaginary" rows and columns.

$$\begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix} *^{-1} \begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix} = 0 \begin{pmatrix} 0 & 1 & - & - \\ 2 & 3 & - & - \\ - & - & - & - \\ - & - & - & - \end{pmatrix} + 1 \begin{pmatrix} - & - & 0 & 1 \\ - & - & 2 & 3 \\ - & - & - & - \\ - & - & - & - \end{pmatrix} + 2 \begin{pmatrix} - & - & - & - \\ - & - & - & - \\ 0 & 1 & - & - \\ 2 & 3 & - & - \end{pmatrix} \\
 + 3 \begin{pmatrix} - & - & - & - \\ - & - & - & - \\ - & - & 0 & 1 \\ - & - & 2 & 3 \end{pmatrix} = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 3 \\ 0 & 2 & 0 & 3 \\ 4 & 6 & 6 & 9 \end{pmatrix}$$

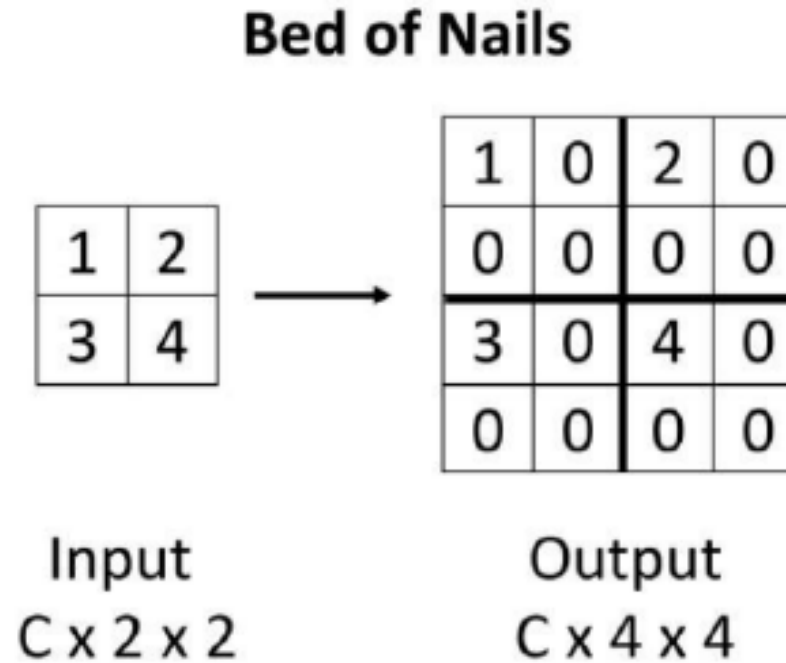
Stride 2 for the previous example.



UPSAMPLING METHODS



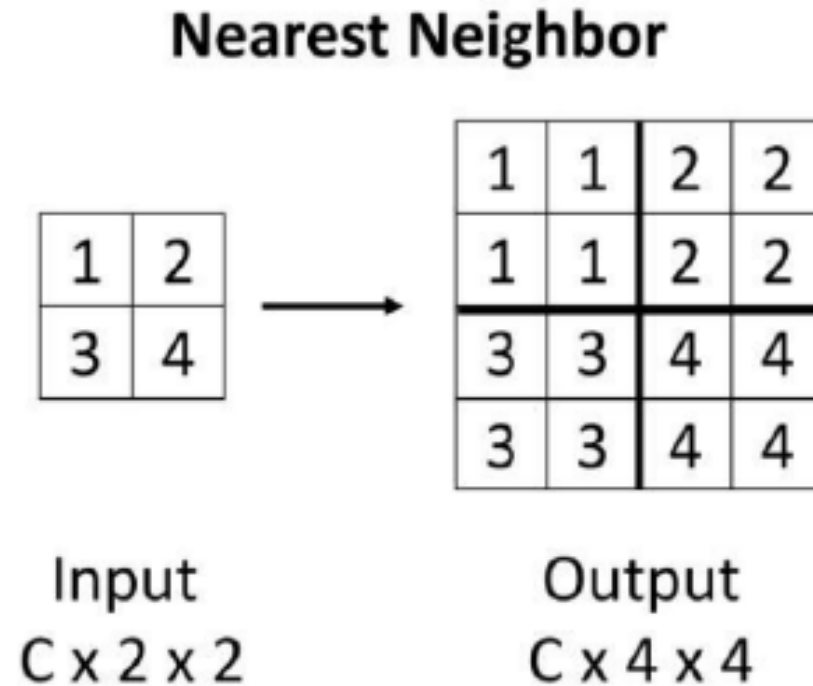
BED OF NAILS



Just fill with 0s!



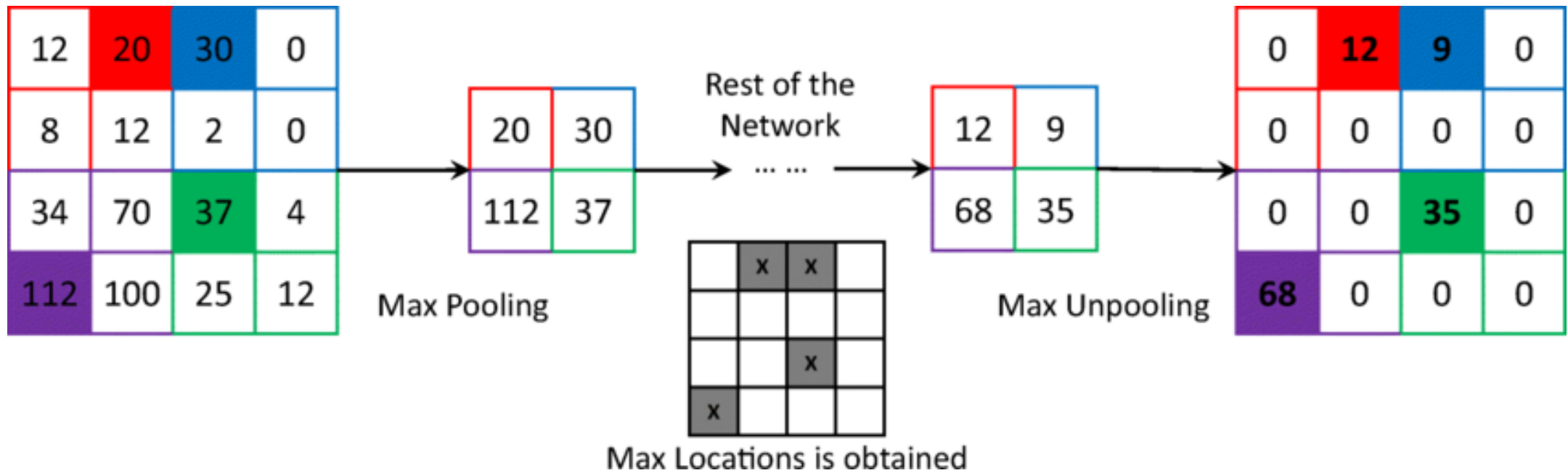
NEAREST NEIGHBOR



Just fill with the same value!



MAX UNPOOLING



WEIGHT INITIALIZATION

Upsamplig from 2 x 2 to 4 x 4 (using stride 2):

Bed of Nails:

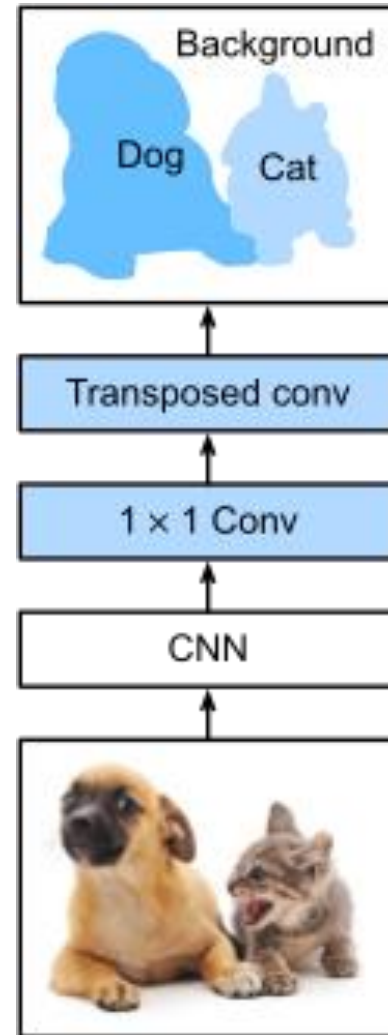
$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} *^{-1} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} a & 0 & b & 0 \\ 0 & 0 & 0 & 0 \\ c & 0 & d & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Nearest Neighbor:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} *^{-1} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} a & a & b & b \\ a & a & b & b \\ c & c & d & d \\ c & c & d & d \end{pmatrix}$$



FCN

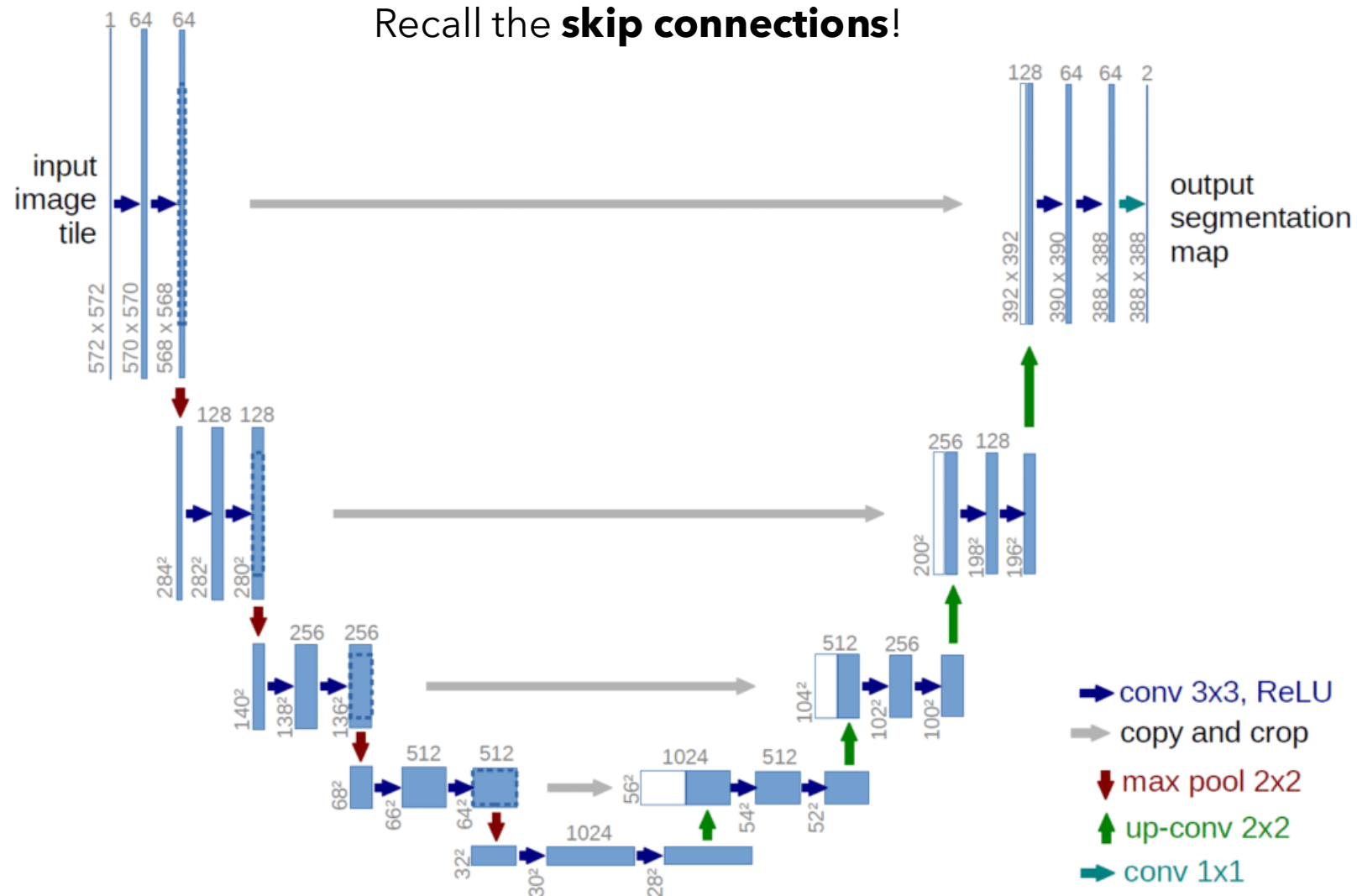


To restore initial spatial dimensions

To obtain one channel per class



U-NET



2. STYLE TRANSFER

DEFINITION OF THE PROBLEM



Content Image

+



Style Image

=

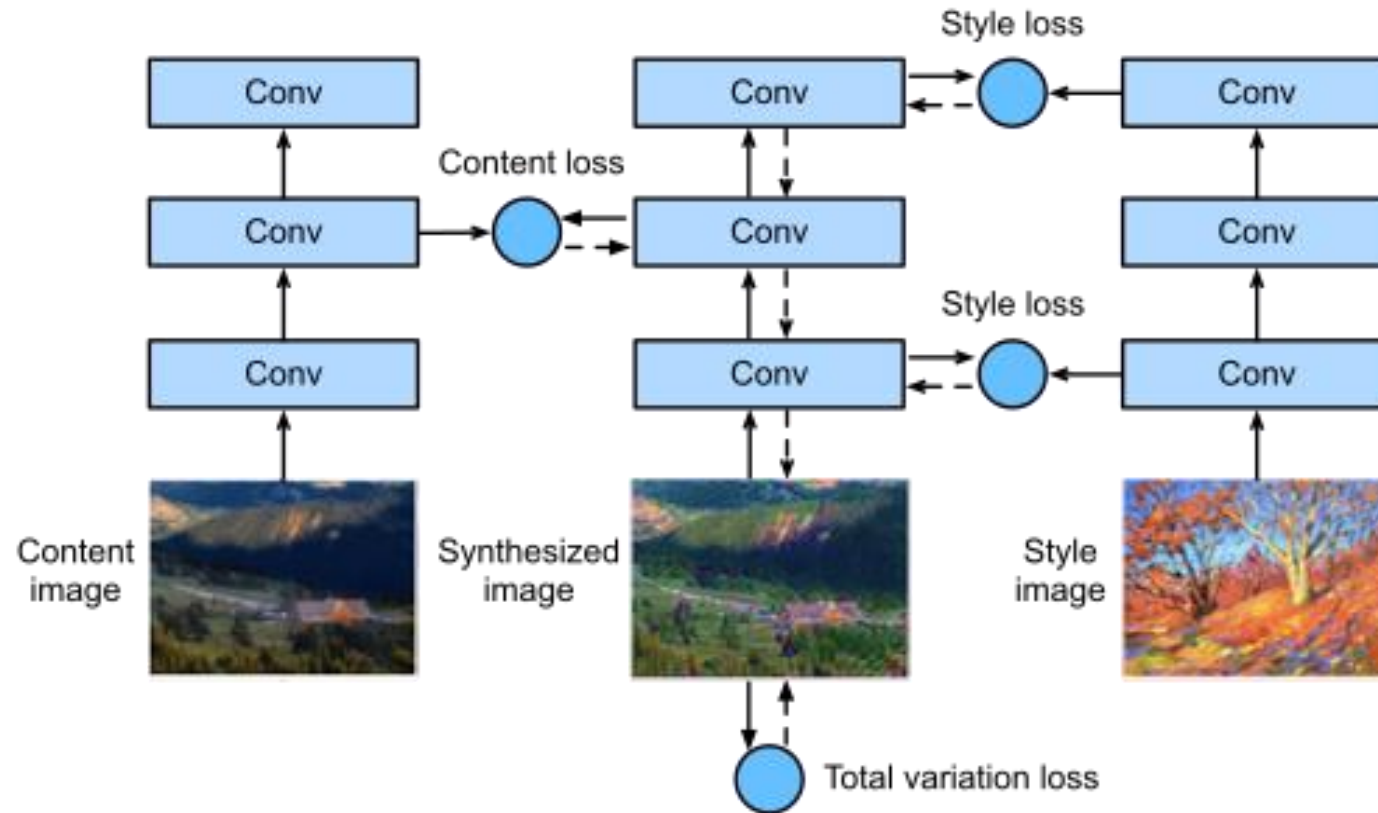


Neural Style Transfer



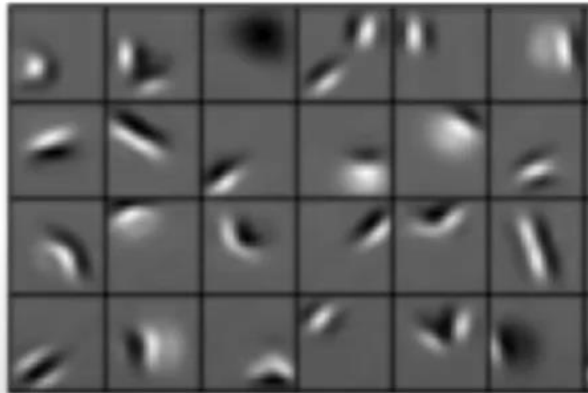
METHOD

- The only **trainable parameters** are the pixels of the synthesized image.
- The synthesized image can be **initialized** in many ways.



EXTRACTING FEATURES

Low level features



Edges, dark spots

Mid level features



Eyes, ears, nose

High level features



Facial structure

- The closer to the **input** layer, the easier to extract **details** of the image.
- The closer to the **output** layer, the easier to extract the **global information** of the image.

EXTRACTING FEATURES

Therefore, in the method, usually:

- Layers chosen for content loss are one or two deep layers (near the output).
- Layers chosen for the style loss are some equally spaced from the input to the output.



CONTENT LOSS

$$\mathcal{L}_C = \underbrace{\frac{1}{n_C} \sum_{r=1}^{n_C}}_{\text{for each feature map}} \underbrace{\frac{1}{C_r H_r W_r} \sum_{c=1}^{C_r} \sum_{i=1}^{H_r} \sum_{j=1}^{W_r} \left(f_r(x)_{cij} - f_r(x^C)_{cij} \right)^2}_{\text{mean of the squared difference of the elements}}$$

Idea: Make synthesized image similar to content image!



STYLE LOSS

$$\mathcal{L}_S = \underbrace{\frac{1}{n_S} \sum_{r=1}^{n_S}}_{\text{for each Gram matrix}} \underbrace{\frac{1}{C_r^2} \sum_{i=1}^{C_r} \sum_{j=1}^{C_r}}_{\text{mean of the squared difference of the elements}} \left((G_r)_{ij} - (G_r^S)_{ij} \right)^2$$

for each Gram matrix

mean of the squared difference of the elements

Gram matrix represents the correlation of the style features of channels i and j .

Idea: Make the style of the synthesized image similar to the style image!



GRAM MATRIX

It represents the **correlation of the style features** of channels i and j . We use this Gram matrix to represent the style output of any style layer.

Feature map: $\begin{pmatrix} x_{111} & \dots & x_{11w} \\ \vdots & \ddots & \vdots \\ x_{1h1} & \dots & x_{1hw} \end{pmatrix}, \dots, \begin{pmatrix} x_{c11} & \dots & x_{c1w} \\ \vdots & \ddots & \vdots \\ x_{chw} & \dots & x_{chw} \end{pmatrix}$ Dimension $h \times w$ (each one) and there are c channels

Flatten it: $\mathbf{X} = \begin{pmatrix} x_{111} & \dots & x_{11w} & \dots & x_{1h1} & \dots & x_{1hw} \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{c11} & \dots & x_{c1w} & \dots & x_{chw} & \dots & x_{chw} \end{pmatrix} = \begin{pmatrix} \mathbf{x}_1 \\ \vdots \\ \mathbf{x}_c \end{pmatrix}$ Dimension $c \times hw$

Gram matrix: $G = \frac{1}{chw} \mathbf{X} \mathbf{X}^\top = \begin{pmatrix} \mathbf{x}_1 \cdot \mathbf{x}_1 & \dots & \mathbf{x}_1 \cdot \mathbf{x}_c \\ \vdots & \ddots & \vdots \\ \mathbf{x}_c \cdot \mathbf{x}_1 & \dots & \mathbf{x}_c \cdot \mathbf{x}_c \end{pmatrix}$ Dimension $c \times c$



TOTAL VARIATION LOSS

$$\mathcal{L}_V = \frac{1}{N} \left(\sum_{c=1}^C \sum_{i=1}^H \sum_{j=1}^{W-1} |x_{c,i,j+1} - x_{c,i,j}| + \sum_{c=1}^C \sum_{i=1}^{H-1} \sum_{j=1}^W |x_{c,i+1,j} - x_{c,i,j}| \right)$$

$$N = C(H(W - 1) + (H - 1)W)$$

Idea: Reduce the noise of the synthesized image!



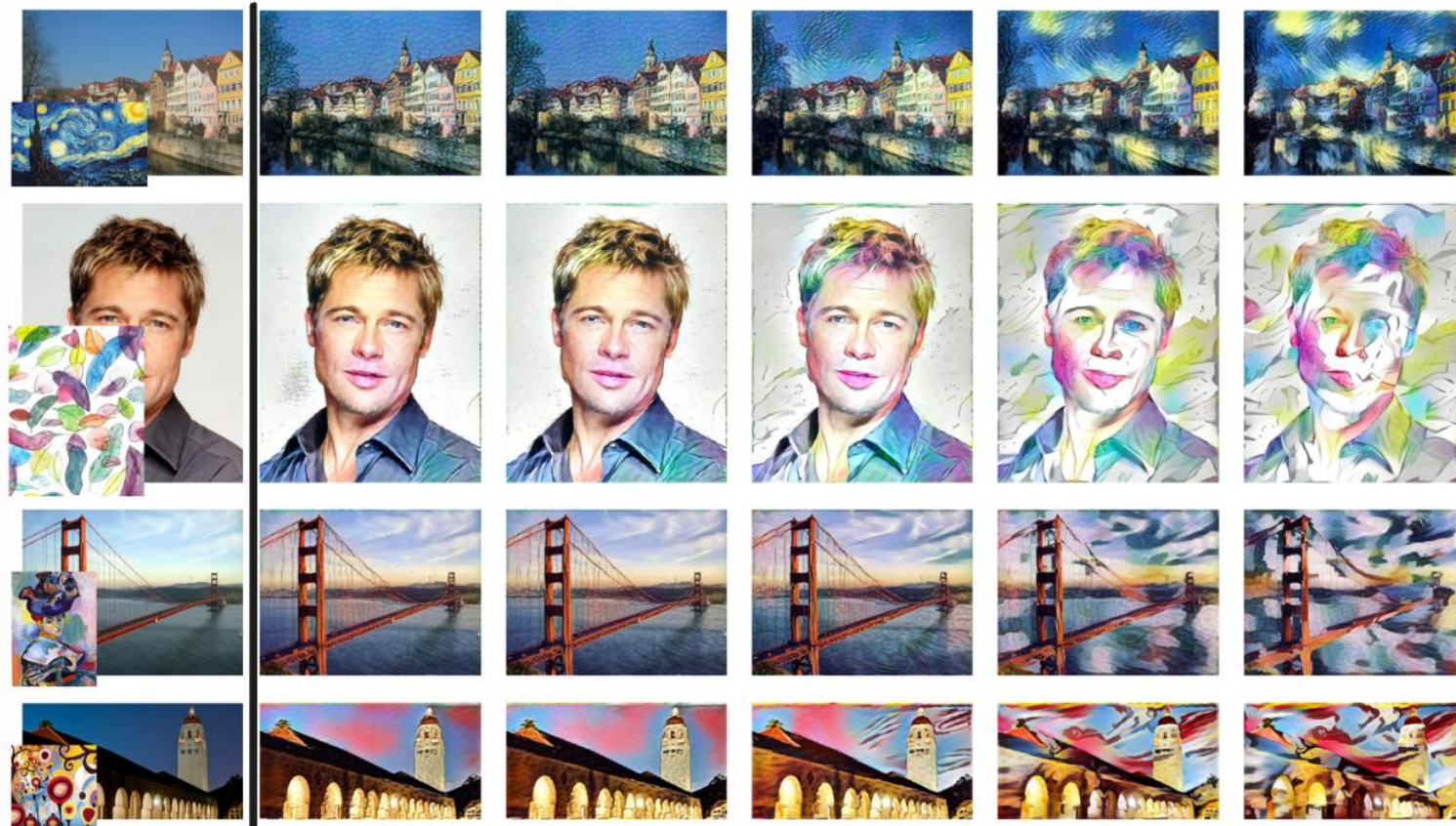
FINAL LOSS FUNCTION

$$\mathcal{L} = \alpha \mathcal{L}_C + \beta \mathcal{L}_S + \gamma \mathcal{L}_V$$

Hint: Optionally we can force $\alpha + \beta + \gamma = 1$ to have a sense of proportion we give to each part.



FINAL LOSS FUNCTION



Content
and Style
Images

←-----→
More weight to
content loss More weight to
style loss



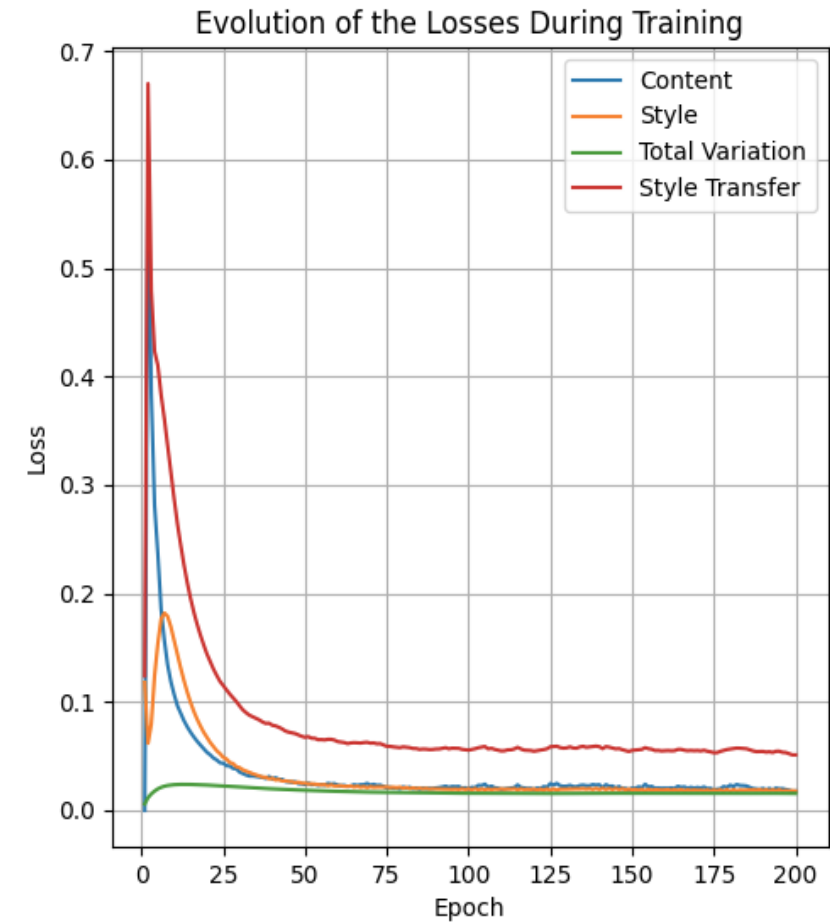
IT IS NOT SO DIFFICULT



Content Image



Style Image



Loss Function

IT IS NOT SO DIFFICULT



1



40



80



200



3 . O T H E R T A S K S

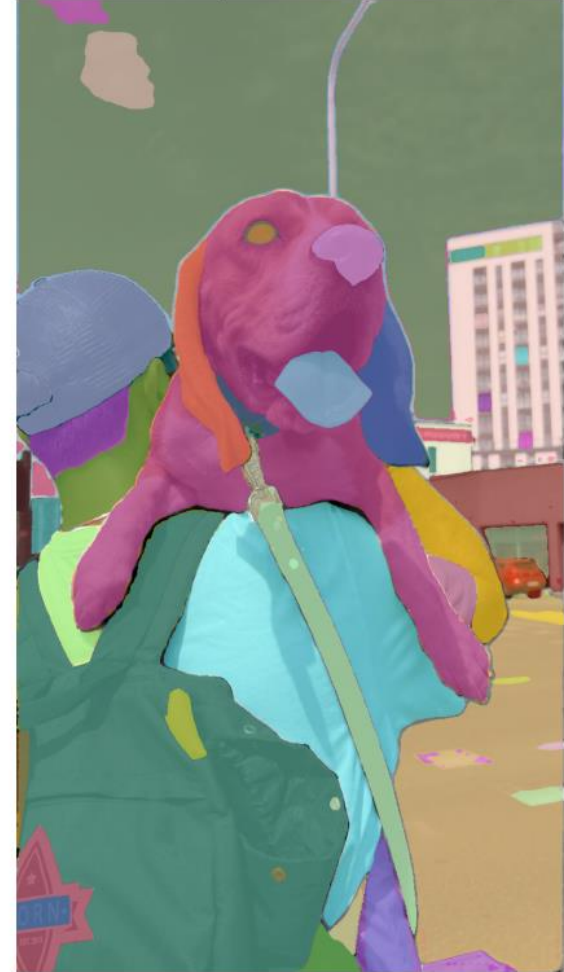
IMAGE SEGMENTATION

- We obtain **regions**, not objects
- **Unsupervised** learning

source image



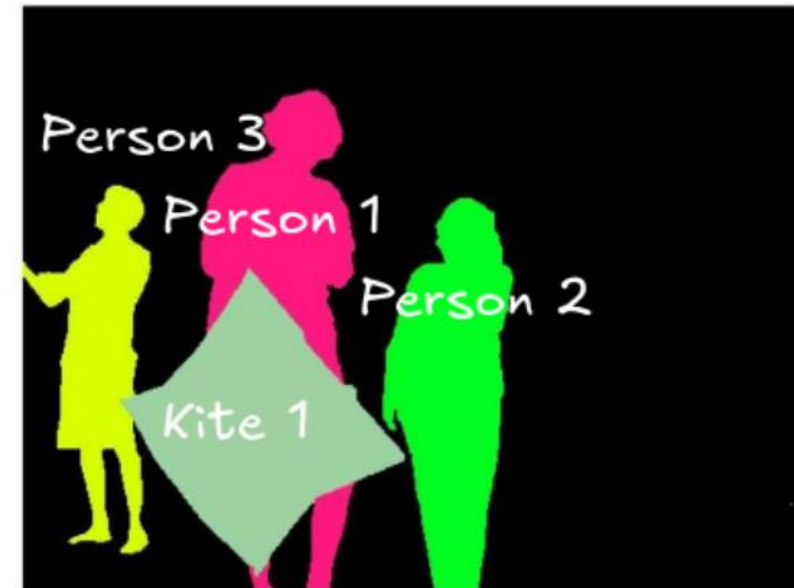
segmented image



INSTANCE SEGMENTATION

Like semantic segmentation, but:

- We also difference between objects of the same class.
- We do not care about "non countable" or "stuff" objects.

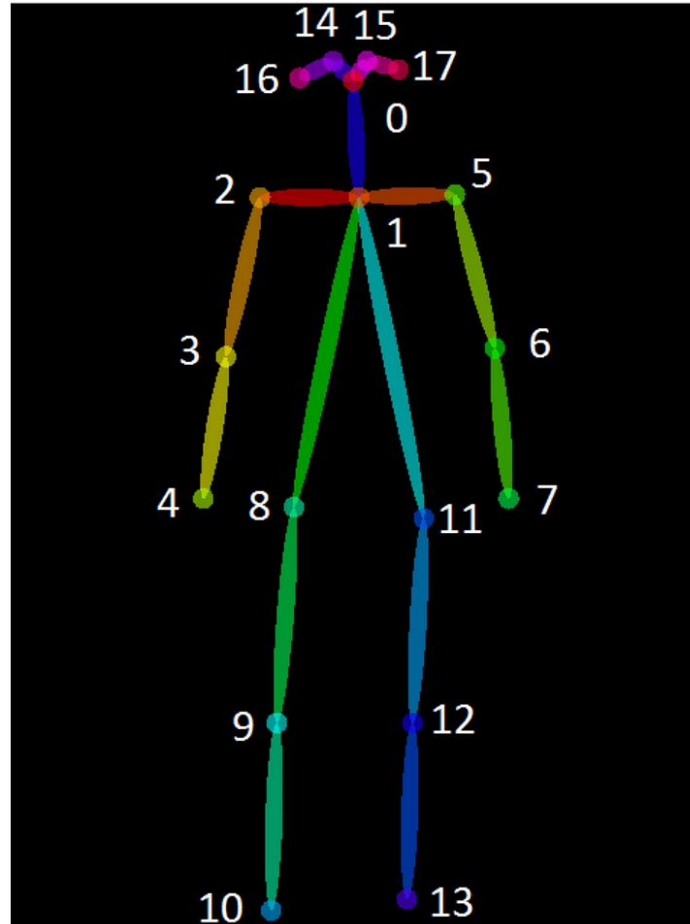


PANOPTIC SEGMENTATION

It is the combination of semantic segmentation and instance segmentation.



KEYPOINT ESTIMATION



T H A N K S !

